

Salesforce CRM Project Documentation

WhatNext Vision Motors: Shaping the Future of Mobility

Project Overview

The WhatNext Vision Motors Salesforce CRM project was developed to streamline vehicle ordering and customer management in the automotive industry. The CRM system centralizes and automates processes such as:

- Vehicle order tracking
- Real-time stock validation
- Test drive scheduling
- Smart dealer assignment

Customers can place orders only when stock is available, with orders automatically routed to the nearest authorized dealer.

Project Objectives

Enhance Customer Experience

- Orders allowed only for in-stock vehicles
- Prevents delays and builds trust

Smart Dealer Assignment

- Orders assigned based on customer location
- Improves service speed and routing accuracy

Improve Operational Efficiency

- Automation reduces manual effort
- Faster order fulfillment and decision-making

Phase 1: Requirement Analysis & Planning

Understanding Business Needs

A centralized CRM was needed to manage:

- Vehicles, dealers, customers
- Orders, test drives, service requests
- Automated communications and stock validations

Project Scope

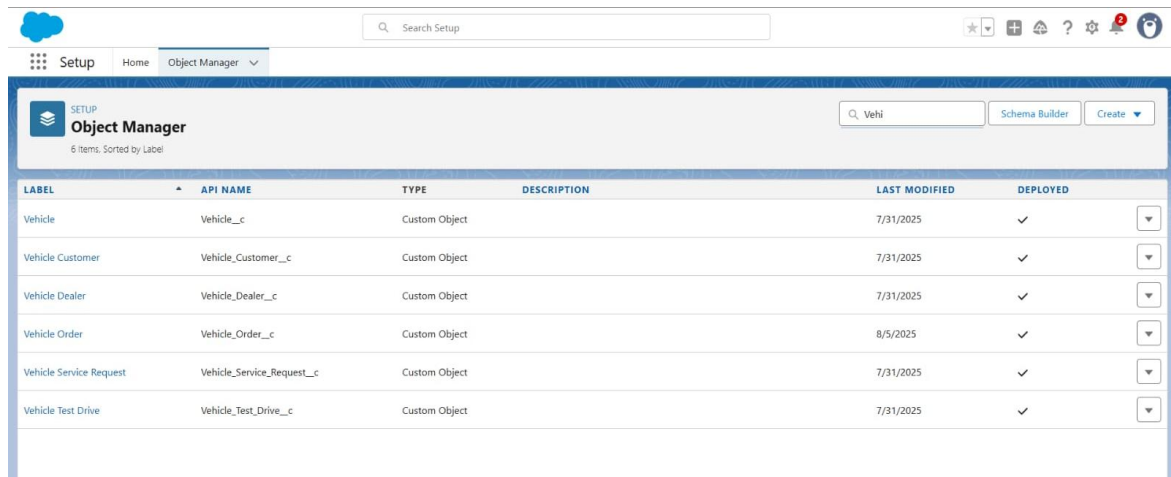
Build a Salesforce CRM that:

- Stores and manages vehicle and dealer information
- Tracks orders and test drives
- Automatically validates stock
- Sends test drive reminders
- Assigns orders to nearby dealers

Data Model Design

Custom objects:

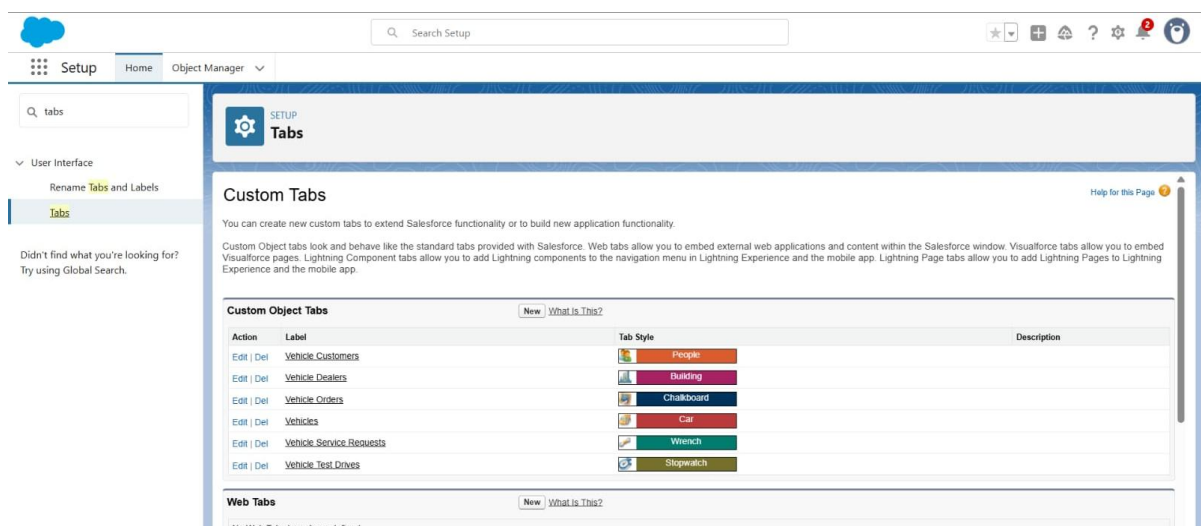
- Vehicle__c
- Vehicle_Dealer__c
- Vehicle_Customer__c
- Vehicle_Order__c
- Vehicle_Test_Drive__c
- Vehicle_Service_Request__c



The screenshot shows the Salesforce Object Manager interface. At the top, there's a search bar with 'Search Setup' and a navigation menu with 'Setup', 'Home', and 'Object Manager'. Below the navigation bar, the 'Object Manager' section is active, showing a search bar with 'Vehi', a 'Schema Builder' button, and a 'Create' button. The main content area displays a table of custom objects.

LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED	DEPLOYED
Vehicle	Vehicle_c	Custom Object		7/31/2025	✓
Vehicle Customer	Vehicle_Customer_c	Custom Object		7/31/2025	✓
Vehicle Dealer	Vehicle_Dealer_c	Custom Object		7/31/2025	✓
Vehicle Order	Vehicle_Order_c	Custom Object		8/5/2025	✓
Vehicle Service Request	Vehicle_Service_Request_c	Custom Object		7/31/2025	✓
Vehicle Test Drive	Vehicle_Test_Drive_c	Custom Object		7/31/2025	✓

Fig 1 Custom Objects



The screenshot shows the Salesforce 'Custom Tabs' configuration page. The left sidebar has a search bar with 'tabs' and a navigation menu with 'User Interface', 'Rename Tabs and Labels', and 'Tabs'. The main content area is titled 'Custom Tabs' and includes a 'Help for this Page' link. Below the title, there's a section for 'Custom Object Tabs' with a 'New' button and a 'What is This?' link. The table below shows the configuration for various custom object tabs.

Action	Label	Tab Style	Description
Edit Del	Vehicle Customers	People	
Edit Del	Vehicle Dealers	Building	
Edit Del	Vehicle Orders	Chalkboard	
Edit Del	Vehicles	Car	
Edit Del	Vehicle Service Requests	Wrench	
Edit Del	Vehicle Test Drives	Stopwatch	

Below the 'Custom Object Tabs' section, there's a 'Web Tabs' section with a 'New' button and a 'What is This?' link. The text below indicates 'No Web Tabs have been defined'.

Fig 2 Custom Tabs

Phase 2: Salesforce Development – Backend & Configurations

Development Environment

- Used Salesforce Developer Org
- Version control via GitHub
- Deployed using Change Sets

Custom Objects & Relationships

- **Vehicle** → stock, model, price
- **Dealer** → location, inventory
- **Customer** → address, contact
- **Order** → linked to vehicle, dealer, and customer

The screenshot shows a web application interface for 'WhatNext Vision M...'. The top navigation bar includes a search bar and several menu items: 'Vehicle Customers', 'Vehicle Dealers', 'Vehicle Orders', 'Vehicle Test Drives', 'Vehicle Service Requests', and 'Vehicles'. The main content area displays a 'Vehicle Customer' record for 'Harshini'. The record is divided into 'Related' and 'Details' tabs. The 'Details' tab is active, showing fields for Customer Name, Email, Phone, Address, Preferred Vehicle Type, and Created By. The 'Owner' field is also visible, showing 'Harshini Kothakonda'. The record was created by 'Harshini Kothakonda' on 8/5/2025 at 10:50 PM and last modified by 'Harshini Kothakonda' on 8/6/2025 at 8:17 AM.

Field	Value
Customer Name	Harshini
Customer Name	Harshini
Email	kothakondaharshini02@gmail.com
Phone	001-807-456-3456
Address	Hyderabad, India
Preferred Vehicle Type	Sedan
Created By	Harshini Kothakonda, 8/5/2025, 10:50 PM
Last Modified By	Harshini Kothakonda, 8/6/2025, 8:17 AM

Fig 3 Vehicle Customer Sample Record

The screenshot shows a web application interface for 'WhatNext Vision M...'. The top navigation bar includes a search bar and several menu items: 'Vehicle Customers', 'Vehicle Dealers', 'Vehicle Orders', 'Vehicle Test Drives', 'Vehicle Service Requests', and 'Vehicles'. The main content area displays a 'Vehicle' record for 'Four Wheeler'. The record is divided into 'Related' and 'Details' tabs. The 'Details' tab is active, showing fields for Vehicle Name, Vehicle Model, Stock Quantity, Vehicle Dealer, Price, Status, and Created By. The 'Owner' field is also visible, showing 'Harshini Kothakonda'. The record was created by 'Harshini Kothakonda' on 8/5/2025 at 10:55 PM and last modified by 'Harshini Kothakonda' on 8/6/2025 at 9:40 AM.

Field	Value
Vehicle Name	Four Wheeler
Vehicle Model	Sedan
Stock Quantity	5
Vehicle Dealer	
Price	\$4,000.00
Status	Available
Created By	Harshini Kothakonda, 8/5/2025, 10:55 PM
Last Modified By	Harshini Kothakonda, 8/6/2025, 9:40 AM

Fig 4 Vehicle Sample Record

Automation with Flows

1. Auto-Assign Dealer Flow

- Trigger: New vehicle order
- Logic: Find customer city → assign nearest dealer
- Updates: Vehicle_Order__c.Dealer__c

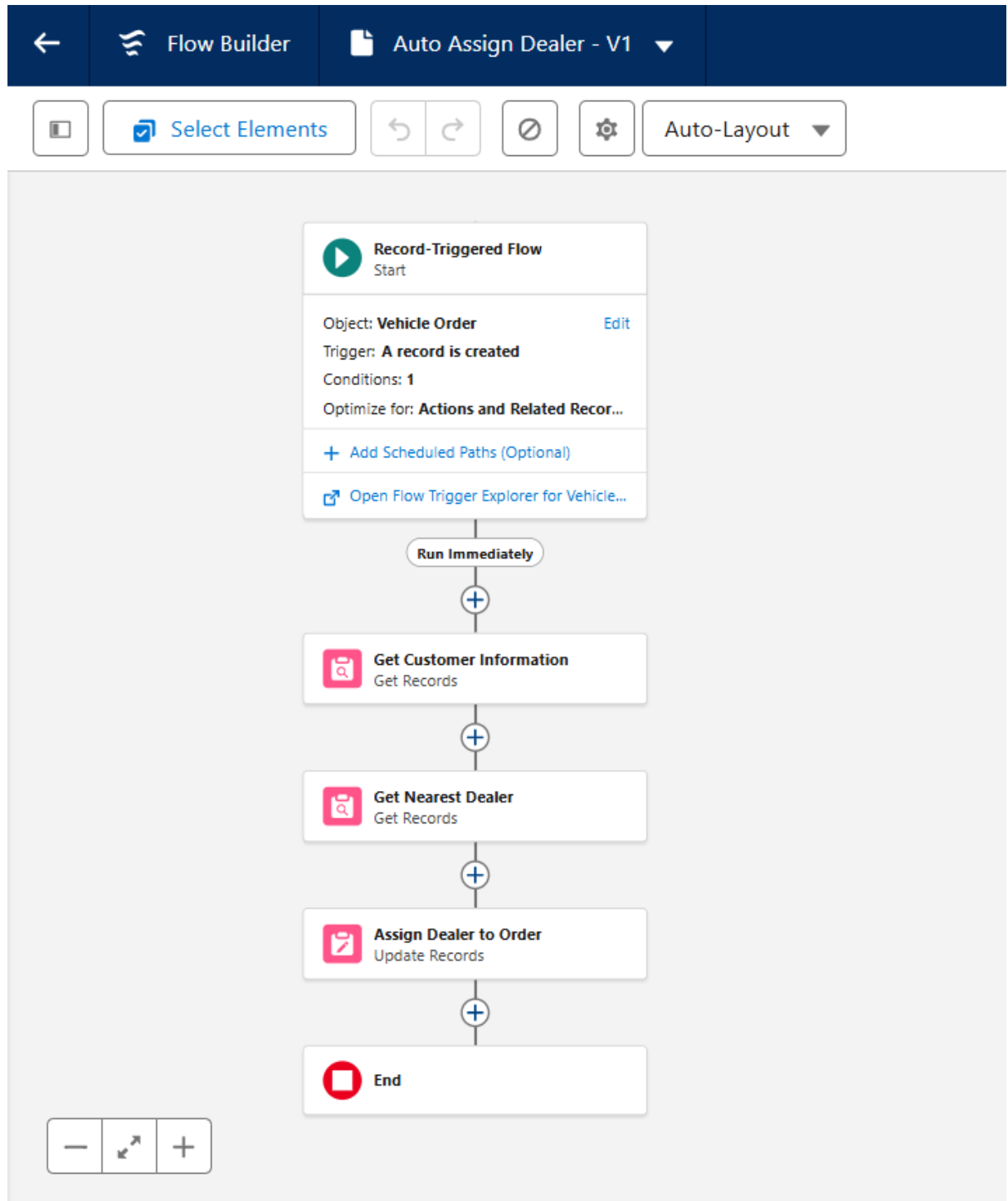


Fig 5 Auto Assign Dealer Flow

2. Test Drive Reminder Flow

- Trigger: Scheduled test drive
- Logic: 1 day before drive → send email
- Output: Reduces missed appointments

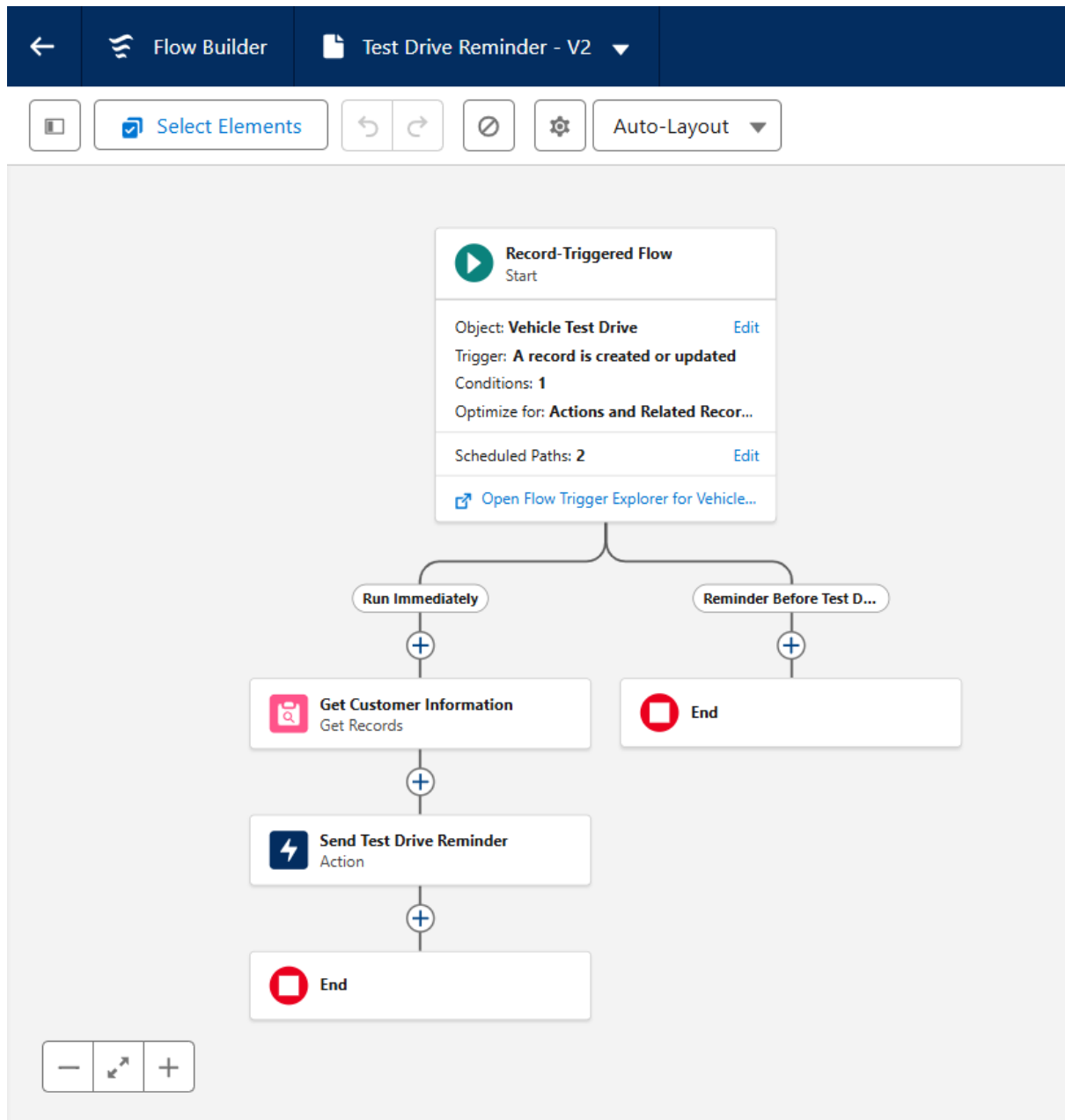


Fig 6 Test Drive Reminder Flow

Working of Test Drive Reminder Flow:

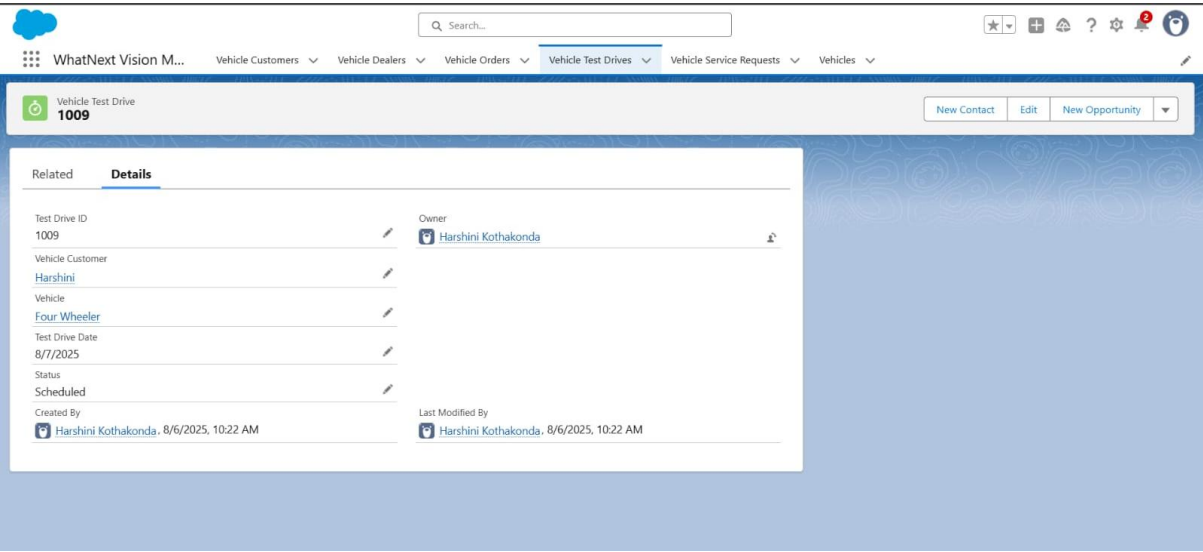


Fig 7 Scheduled Test Drive Record

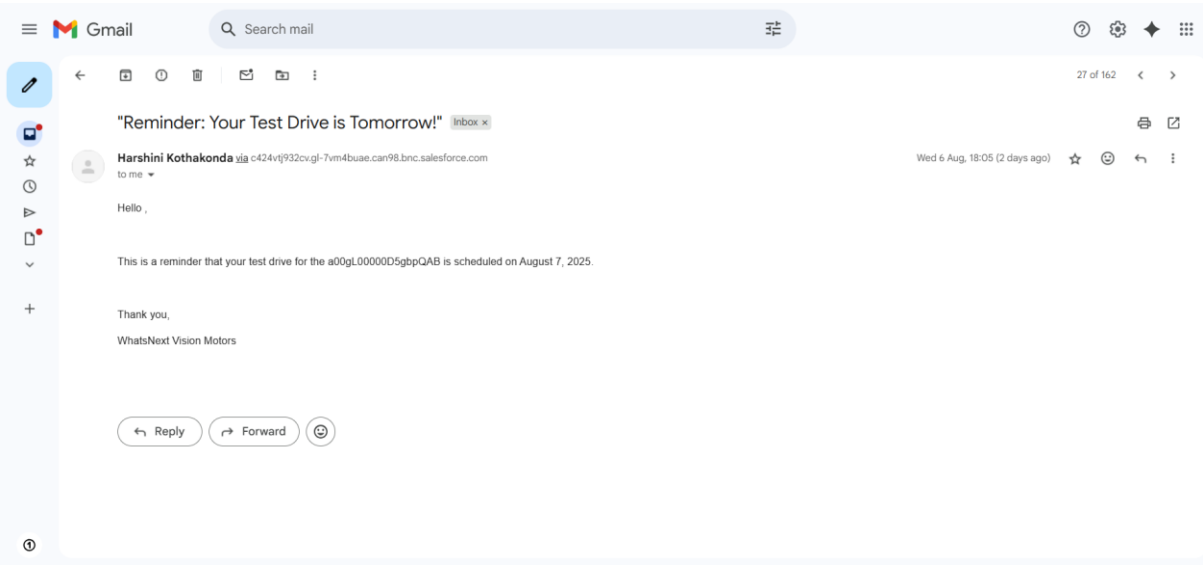


Fig 8 Reminder Mail

This is an automated reminder email triggered from Salesforce when a test drive status is set to "Scheduled." The email is sent to the customer 1 day before the scheduled test drive date. It includes the test drive date and is linked to the relevant Test Drive record (e.g., a00gL00000D5gbQAB) in Salesforce. The message is sent on behalf of the WhatsNext Vision Motors to confirm the appointment and ensure customer preparedness.

Apex Development

Apex Trigger Handler: VehicleOrderTriggerHandler

- Prevents orders if stock = 0
- Reduces stock when confirmed
- Follows trigger best practices

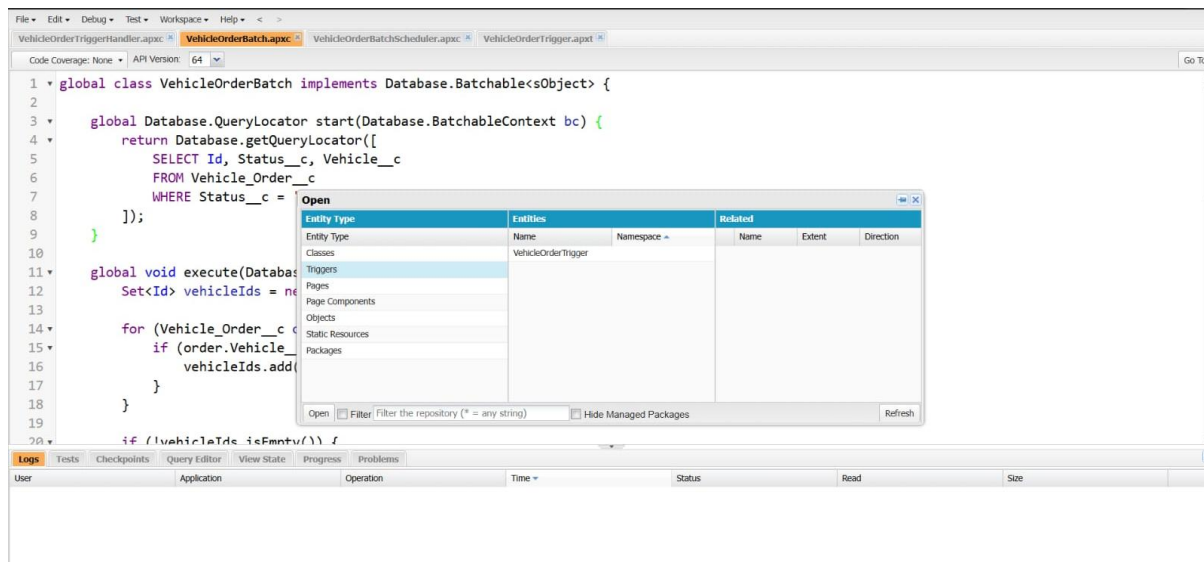


Fig 9 Vehicle Order Trigger

Trigger Class: VehicleOrderTrigger

- Calls the handler during insert/update
- Separates logic from trigger

Batch Apex: VehicleOrderBatch

- Processes all orders with status "Pending"
- If stock available → mark as "Confirmed", reduce stock

Scheduler Class: VehicleOrderBatchScheduler

- Runs the batch daily (e.g., 12 PM)
- Ensures automatic order updates

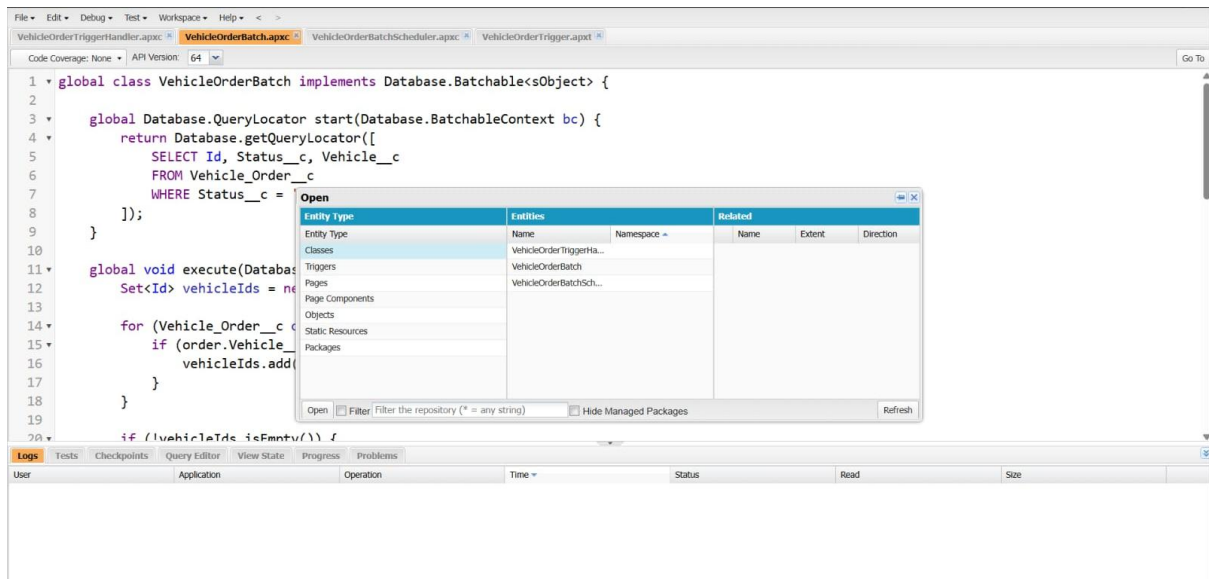


Fig 10 Apex Classes

Phase 3: UI/UX Development

Lightning App Setup

- App name: "WhatNext Vision Motors"
- Tabs: Vehicles, Dealers, Orders, Test Drives, etc.
-

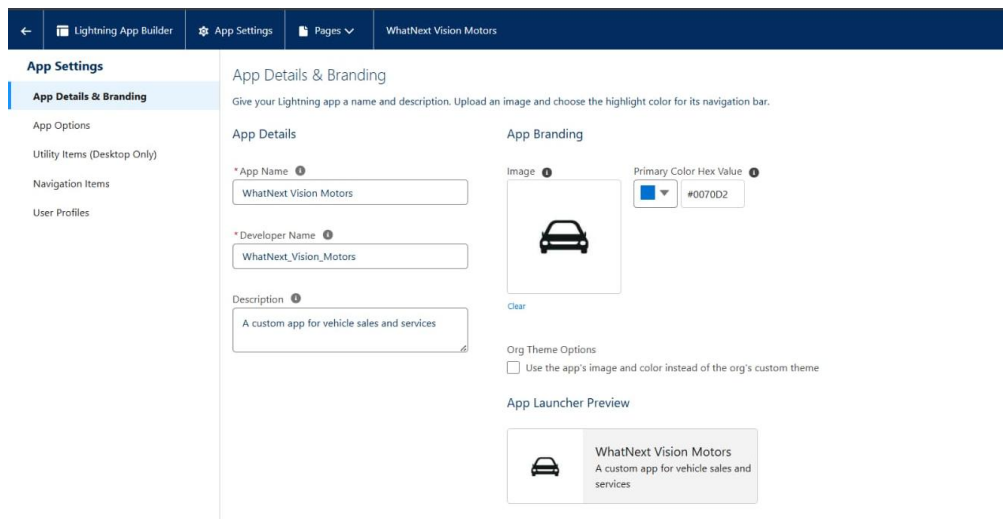


Fig 11 Lightning App

Page Layouts & Dynamic Forms

- Conditional field visibility
- Clean layout for key objects

User Access & Permissions

- Standard profiles + permission sets
- Controlled access to custom objects

Phase 4: Data Migration, Testing & Security.

Python-Scripted Sample Data

To generate realistic and structured test data, a custom Python script was created. This script produced:

- Vehicle records (with varying stock levels and models)
- Dealer entries based on different cities
- Sample customer profiles with addresses and emails

This helped simulate real-world scenarios during automation and batch testing.

Data Import Wizard

To simplify initial data population for testing and validation, the Salesforce Data Import Wizard was used. It enabled importing data into custom objects such as:

- Vehicle__c
- Vehicle_Dealer__c
- Vehicle_Customer__c

This tool allowed easy mapping of CSV columns to Salesforce fields and helped validate records before bulk creation.

The screenshot shows the Salesforce Setup page for Bulk Data Load Jobs. The interface includes a search bar, navigation tabs (Setup, Home, Object Manager), and a sidebar with a search for 'data import' and a link to the 'Data Import Wizard'. The main content area displays the details of a job submitted by Harshini Kothakonda. The job is for an 'Insert' operation on the 'Vehicle Dealer' object, using a CSV file. The progress is 100%, and 10 records were processed successfully. A table at the bottom lists the batches, showing a single batch with ID 751gL000008cP2R, processed on 8/6/2025 at 9:53 AM, with a total processing time of 86 ms and 31 API active processing time.

Submitted By	Operation	Insert	Total Processing Time (ms)
Harshini Kothakonda	Queued Batches	0	31
Start Time: 8/6/2025, 9:53 AM PST	In Progress Batches	0	0
End Time: 8/6/2025, 9:53 AM PST	Completed Batches	1	0
Time to Complete (hh:mm:ss)	Failed Batches	0	0
Object: Vehicle Dealer	Progress	100%	
External ID Field	Records Processed	10	
Content Type: CSV	Records Failed	0	
Concurrency Mode: Parallel	Retries	0	
API Version: 64.0			

View Request	View Result	Batch ID	Start Time	End Time	Total Processing Time (ms)	API Active Processing Time (ms)	Apex Processing Time (ms)	Records Processed	Records Failed	Retry Count	State Message	Status
View Request	View Result	751gL000008cP2R	8/6/2025, 9:53 AM	8/6/2025, 9:53 AM	86	31	0	10	0	0	Completed	Completed

Fig 12 Data Import

Testing Overview

Scenarios Covered:

- Vehicle creation and validation
- Order placement (confirmed/pending)
- Stock update triggers
- Batch job logic
- Test drive reminders

Phase 5: Deployment & Maintenance

Deployment Strategy

Used Change Sets to move metadata including:

- Custom objects/fields
- Validation rules
- Apex classes & triggers
- Record-triggered flows

System Maintenance

- Monitored batch jobs via Salesforce Scheduled Jobs
- Used debug logs for troubleshooting

Conclusion

The Salesforce CRM implementation modernized WhatNext Vision Motors' sales and service operations. Key wins include:

- Automation of stock validation and order routing
- Improved customer satisfaction
- Reduced manual effort
- Scalable foundation for future upgrades

Future Enhancements

1. **Chatbot Integration**
Implement Einstein Bot for 24/7 customer queries, test drive booking, and order status.
2. **Advanced Reporting**
Use Einstein Analytics for performance dashboards and predictive insights.
3. **Service Request Automation**
Automate Vehicle_Service_Request__c processes: assignments, reminders, and history tracking.